SG No. 106 C_{4v}^{8} $P4_2bc$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `4a`, site symmetry: `2..`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[0,\ 0,\ z]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,-4]`
3	$[X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,-6]`
4	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,-8]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,4]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,6]`
4	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[0,\ 0,\ z]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4]`
5	$[X,\ -Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5]`
6	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[6]`
7	$[-Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7]`
8	$[Y,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[8]`

* Wyckoff site: `4b`, site symmetry: `2..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[3,-4]`
3	$[X,\ -Y,\ 0]$	$\left[\frac{1}{2},\ 0,\ z\right]$	`[5,-6]`
4	$[-Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,-8]`

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`

continued ...

Table 5

No.	vector	center	mapping
2	$[0,\, 0,\, Z]$	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[3,4]`
3	$[0,\, 0,\, Z]$	$\left[\frac{1}{2},\, 0,\, z\right]$	`[5,6]`
4	$[0,\, 0,\, Z]$	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[7,8]`

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[0,\, \frac{1}{2},\, z\right]$	`[1]`
2	$[-X,\, -Y,\, Z]$	$\left[0,\, \frac{1}{2},\, z\right]$	`[2]`
3	$[-Y,\, X,\, Z]$	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\, -X,\, Z]$	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[4]`
5	$[X,\, -Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, z\right]$	`[5]`
6	$[-X,\, Y,\, Z]$	$\left[\frac{1}{2},\, 0,\, z\right]$	`[6]`
7	$[-Y,\, -X,\, Z]$	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[7]`
8	$[Y,\, X,\, Z]$	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[8]`

* Wyckoff site: `8c`, site symmetry: `1`

Table 7: Wyckoff bond: `8a@8c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	`[1]`
2	$[-X,\, -Y,\, Z]$	$[-x,\, -y,\, z]$	`[2]`
3	$[-Y,\, X,\, Z]$	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\, -X,\, Z]$	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	`[4]`
5	$[X,\, -Y,\, Z]$	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	`[5]`
6	$[-X,\, Y,\, Z]$	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	`[6]`
7	$[-Y,\, -X,\, Z]$	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[7]`
8	$[Y,\, X,\, Z]$	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[8]`